Daria Sorokina

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Applied ML/AI Scientist with 15+ years of industry experience, combining hands-on development of state-of-the-art ML/AI models with a track record of leading large-scale, high-impact ML initiatives. Recent work at Shopify focuses on LLM training, knowledge distillation, and generative retrieval. At Amazon Search, led cross-organizational ML initiatives adopted across diverse product categories and global marketplaces, delivering multi-billion-dollar business impact.

Education

2004 – 2008 | PhD in Computer Science, Cornell University. Advised by Rich Caruana.
1998 – 2003 | Diplom with Honors in Applied Mathematics and Computer Science, Moscow State University.

Work experience

2026 | **Senior Staff Machine Learning Engineer, Foundational Models, Shopify**

Owned knowledge distillation for SemanticID-LLM generative retrieval in Shopify Search. Worked hands-on on continued pre-training (CPT), supervised fine-tuning (SFT), and distillation of student models, targeting production-scale inference. Implemented and experimented with on-policy and off-policy distillation, hard/soft labels, cross-tokenizer distillation, forward KL and Jensen-Shannon losses. Through deep dives into teacher/student model quality gaps, identified poor coverage of rare brand names as a key limitation and drove changes to the CPT data mixture to address it, improving student model quality. Owned offline evaluation and built reproducible ML pipelines for business and relevance metrics across retrieval and ranking.

Contributed to Shopify's Unified Distillation Platform (UDP) and Tangle ML pipeline ecosystem. Was a member of the team tasked with building standardized distillation pipelines across Shopify use cases. Developed ML training and distillation pipelines, integrated multi-node distributed training, and implemented checkpointing to Hugging Face and Google Cloud Storage (GCS), including asynchronous checkpointing.

Collaborated with Liquid AI and Shopify's Sidekick team on a **mixture-of-experts (MoE) model with context parallelism** for Sidekick Suggested Actions, an LLM task relying heavily on tool calls: built the UDP training pipeline, diagnosed issues in the model training code, and drove technical coordination across three teams. Presented biweekly progress updates to the CTO and senior leadership.

Extensively used AI-assisted development with Cursor and Codex in day-to-day research and engineering. Built agents to automate workflows such as autonomous ML pipeline monitoring, failure diagnosis and recovery, and PR review/CI/merge cycles; optimized for token efficiency without sacrificing quality.

2014 – 2025 | **Science Lead, Search Relevance, Amazon**

Led multiple high-impact ML initiatives for training ranking models across Amazon's search stack. Delivered launches contributing billions in annualized sales. Supervised virtual teams of up to 15 experimenters, ran collaborations across org boundaries, aligning multiple orgs on best practices and reaching consensus among technical leadership. Mentored 20+ scientists with at least 10 advancing to leading projects and 5 achieving L6/L7 promotions. Established and ran a cross-org peer-review program for ranking experiments. Participated in org-wide annual planning and goal setting. Collaborated with finance teams to estimate anticipated increases in operational costs.

Developed algorithms and led a series of cross-team projects for novel multi-objective (MOO) techniques that enabled easy introduction of new objectives for ranking models, gaining VP-level visibility. Our production models, trained on 4-6 diverse objectives (e.g., conversion, brand quality, shipping speed, sales volume), routinely improved online performance on the corresponding metrics, culminating in a record-breaking launch that increased annualized sales by >$1B. Co-authored an internal paper on novel MOO ranking targets. Developed algorithms for automatically tuning MOO coefficients jointly with the model parameters extending Bayesian hyper-parameter optimization with criteria of selection from Pareto frontier. Oversaw internal tool development for these algorithms and drove adoption across 10+ 'away teams' responsible for specialized models (e.g., apparel, books, or international marketplaces like India and Japan), providing mentorship and guidance. Overall, involved in 25+ MOO model launches (11 hands-on); techniques became default for multiple orgs.

Designed and drove adoption of multiple critical model evaluation policies. Overhauled offline evaluation for MOO models, mandating alignment with online guardrail metrics. Authored a launch policy for mid-size marketplaces leveraging interleaving and A/B experiments; it enabled several successful launches. Negotiated policy for model-based vs. manual assessment of search quality metrics. Contributed to org tenets for multi-objective models. Collaborated closely with infra to ensure accurate latency measurements per ranking phase, enhancing monitoring tools for performance risk assessment.

Led conversion/sales optimization project improving training targets and offline evaluation metrics, reducing online-offline gap by 58%. Successfully deployed the new statistical formula, delivering over $500M in annualized sales gains, and drove its adoption as the standard for evaluation metrics, and for training main ranker models. Because offline performance depends on the same formula as the target, comparing formulas solely on offline data is infeasible. To address this, built a generalized framework for iterative target refinement via root-cause analysis and hypothesis testing using interleaving experiments. We ran 6 such experiments with ∼25 treatments each, evaluating 72 target variants.

Co-led development and launch of the deep learning final stage ranker model using our algorithms for MOO, conversion/sales objectives, and feature selection. The model is based on the HydraNet architecture, where each head optimizes for one of the objectives, with an MLP aggregator providing the final optimization on the Pareto front. Demonstrated improvements both in online sales metrics and in qualitative perception of results.

Founded and scaled Search Relevance Model Training Bar Raisers (MTBR) program hands-on for 6 years. Model Bar Raisers provide peer review for model trainers, helping all ranking experimenters train and test high-quality ranking models. MTBR sign-off is required before online experiments. This program drove a cultural shift towards rigorous experimentation and knowledge sharing, elevating the quality of 600 reviewed experiments across numerous teams, including 'away teams' with specialized models.

Led re-architecture initiative to optimize the interplay between fast and main rankers. Designed and executed a comprehensive learning experiment with 25 parameter combinations to evaluate the potential impact of different system configurations before large-scale deployment, effectively mitigating the risk of negative performance regressions.

Invented FirTree (feature-interactions-and-regression-tree), a compact and precise model that optimizes the performance/size tradeoff through an elaborate training algorithm. Led multiple projects that launched FirTree models as first-phase rankers in Amazon's multi-phase search ranking stack, saving millions in operational costs and enabling downstream use of complex models. Other teams successfully adopted FirTree models for diverse use cases: filters for irrelevant search results and priors in behavioral features.

Designed several deep dive methodologies and performed systematic deep dives to root-cause common model performance issues. For instance, in 2024, ran a series of deep dives on broad query performance, leading to four immediate fixes and >$200MM increase in annualized sales. Actively influenced the development of model explanation tools enhancing interpretability for such deep dives.

As the most experienced IC in ML model training within the early-stage Search Relevance org, resolved several critical gaps in our default algorithms. Improved Search's **feature selection** stack: made kitchen sync technique viable for heterogeneous data (published internally); adapted a gradient boosting built-in feature selection algorithm to our tasks (published in AISTATS). Drove the org's first **cold start** initiative, reducing ticket queue by 60% through a full-stack solution: built reporting pipeline, root-caused issues, launched cold-start-aware behavioral features, measured wins. Redesigned "**Sort by Customer Review**" logic, improving UX and driving up sales; formula remains in production 9 years later. Developed **data introspection tools** used to detect data issues, inspire new features, and surface labeling problems (published internally).

Chair of WSDM'19 Industry Day, increased attendance by 2.5x. Delivered MLConf'16 talk on Amazon Search, **2nd most-viewed MLConf talk** (>13K views); earlier version published at SIGIR'16. Authored extensive **internal documentation** (algorithm designs, research papers, tutorials, project proposals/reports) for diverse audiences from peers to leadership.

2012 – 2014 | **Senior Data Scientist at LinkedIn**

Delivered multiple machine learning solutions for search. Built search spam detection from scratch. Created a new algorithm for static (query-independent) ranking of profiles. Designed and trained new models for personalized ranking in job search.

2010 – 2011 | **Relevance Scientist at Yandex Labs, California office of Yandex**

Contributed to numerous projects for English web search, including ranking algorithms, feature evaluation, selection of documents for the search index, feature extraction from user logs, image search, and query suggestions. Won **fourth place in Yahoo! Learning to Rank Challenge.**

2009 – 2010 | **Postdoctoral fellow at Auton Lab, Carnegie Mellon University**

Investigated feature selection approaches for limited high-variance data for the USDA project on Salmonella risk prediction. Participated in other projects such as mobile threat detection with Livermore Labs and shoe demand forecast for ECCO.

2005 – 2008 | **PhD thesis: "Modelling Additive Structure and Detecting Interactions with Groves of Trees"**

Invented Additive Groves, a novel regression ensemble. This algorithm produces a strong prediction model and allows detecting non-additive interactions in large data sets. Its application was part of a joint project with Cornell Lab of Ornithology on tracking environmental change. Built on this work in several successful projects in CMU, LinkedIn, Amazon.

summer 2007 | **Intern at Google Pittsburgh, Search Ads**

Project on fast large-scale feature evaluation in logistic regression; evolved into a chapter in "Scaling Up Machine Learning".

Selected publications and talks

Cuize Han, Nikhil Rao, Daria Sorokina, Karthik Subbian. Scalable Feature Selection for (Multitask) Gradient Boosted Trees. 23rd International Conference on Artificial Intelligence and Statistics (AISTATS'20).

Daria Sorokina, Erick Cantu-Paz. Amazon Search: The Joy of Ranking Products. SIGIR'16 industry track talk.

Daria Sorokina. Recruiters, Job Seekers, and Spammers: Innovations in Job Search at LinkedIn. ECIR'13 workshops keynote.

Jeremy Kubica, Sameer Singh and Daria Sorokina. Parallel large-scale feature selection. Book chapter in Scaling Up Machine Learning: Parallel and Distributed Approaches. Cambridge University Press. 2011.

Daria Sorokina. Application of Additive Groves to the Yahoo! Learning to Rank Challenge. Presented at ICML'10.

Daria Sorokina, Rich Caruana, Mirek Riedewald, Daniel Fink. Detecting Statistical Interactions with Additive Groves of Trees. 25th International Conference on Machine Learning (ICML'08).

Daria Sorokina, Rich Caruana, Mirek Riedewald. Additive Groves of Regression Trees. 18th European Conference on Machine Learning (ECML'07). **Best Student Paper award**.

Daria Sorokina, Johannes Gehrke, Simeon Warner, Paul Ginsparg. Plagiarism Detection in arXiv. IEEE ICDM'06.

R. Caruana, M. Elhawary, A. Munson, M. Riedewald, D. Sorokina, D. Fink, W. Hochachka, S. Kelling. Mining Citizen Science Data to Predict Prevalence of Wild Bird Species. ACM KDD'06.